

START HERE • NEW TECHNICIAN

Fiber Tech Starter Guide

The plain-English basics for your first days on the floor.

Read this first. It keeps things simple:

Stay safe • what fiber is • the plugs you'll touch
TX and RX • keep it clean • read a work order
when a link is down • labeling • words to know

When in doubt, STOP and ask. Nobody expects you to know it all on day one.
Always follow your site's own rules, safety steps, and records.

Welcome!

This little guide is for your **first days** as a fiber technician. It skips the deep theory and just tells you what things are and what to do. Keep it in your bag. Once it all feels easy, move up to the full *L2 Fiber Technician Field Manual*.

THINK OF IT LIKE — What a fiber tech really does

Your job is simple to say: **move light through glass, cleanly, to the right place, and write down what you did.** Everything in this guide supports that one idea.

A few promises to make your life easy:

- **You will not know everything on day one.** That's normal. Ask questions.
- **Slow and correct beats fast and wrong.** A rushed mistake can take a service down.
- **The record is the boss.** If the paperwork and the real world don't match, stop.
- **Clean, then connect.** Dirt is the #1 cause of problems. Every single time.

STEP 1

Stay Safe

Nothing here is worth getting hurt. Learn these few rules and you'll be fine.

STOP — Never look into a fiber or a port

Fiber carries **invisible** laser light. You won't see it or feel it, but it can hurt your eyes. **Never** look into the end of a cable or an open port. To check an end, use a fiber scope — never your eye.

The short safety list

- **Wear safety glasses** when you handle bare fiber.
- **Bits of fiber are like tiny glass splinters.** They're nearly invisible and won't come out. Work over a mat, drop scraps in the little scrap jar, and never touch your face.
- **No food or drink** at the fiber bench — a scrap could end up in it.
- **Ladders and lifts:** only use them if you're trained. Three points of contact, don't overreach.
- **See a lock or tag on something?** Don't touch it. Someone's safety depends on it.
- **Know your exits** and where the eyewash station is before you need them.

DO THIS — If something feels unsafe

Stop. Step back. Ask your lead. You will never get in trouble for pausing to stay safe. That's always the right call.

STEP 2

What Is Fiber, Really?

Fiber is a hair-thin strand of **glass** that carries light instead of electricity. A flash of light on means a 1, off means a 0 — that's how your data travels.

THINK OF IT LIKE — A fiber is like a garden hose for light

Shine a flashlight in one end and it comes out the other — even around gentle bends. Kink the hose (bend it too tight) and the flow chokes off. Same with fiber: **never bend it into a tight loop.**

Two big families

- **Single-mode (SMF)** — tiny glass core, goes really far. Jacket is usually **yellow**. Think long distances.
- **Multimode (MMF)** — bigger core, goes shorter distances but is cheaper. Jacket is usually **aqua** (or orange on older stuff). Think inside-the-building.

DO THIS — The one rule that matters most here

Don't bend fiber too tight and don't crush it. A good habit: never make a loop smaller than a soda can, and never cinch a zip tie down hard on it. Tight bends leak the light out and cause faults.

STEP 3

The Plugs You'll Touch

There are many connectors out there, but as a beginner you'll mostly see just two. Learn these first.

LC — the little one (most common)

- Small connector, usually clipped together in **pairs** (duplex).
- One side sends light, the other side receives it (more on that next step).
- This is on almost everything you'll patch day to day.

MPO / MTP — the wide one (many fibers in one plug)

- A wide, rectangular plug that holds **many** fibers at once (often 8 or 12).
- Used for the big, fast links. Has a little **key** (bump) on one side — it only goes in one way.
- MTP is just a fancy brand of MPO. They plug together.

STOP — Green plugs and blue plugs don't mix

Connector ends are color-coded by how they're polished. **Green (APC)** only goes with green. **Blue (UPC)** only goes with blue. Forcing green into blue ruins both ends. If the colors don't match, stop.

TIP — Push until it clicks

Connectors seat with a gentle **click**. If it doesn't click, it isn't seated — and a loose connector looks connected but won't work. Never force it; line it up and push straight in.

STEP 4

TX and RX (Send and Receive)

Every link needs to **send** in one direction and **receive** in the other. You'll see these two little words everywhere:

- **TX** = Transmit = the light going **out** (sending).
- **RX** = Receive = the light coming **in** (receiving).

THINK OF IT LIKE — Like a two-lane road, or a phone call

You talk on one line, the other person talks on the other. If both cars drove in the same lane, nobody gets anywhere. **TX on one end has to reach RX on the other end.** That crossover is called *polarity*.

What this means for you

- On a duplex LC pair, the two fibers **cross over** so send meets receive. Usually the cable does this for you.
- Sometimes the work order says **ROLL**. That just means: **swap the two fibers** so send and receive line up. Use the jumper or clip that flips them.
- **Weird clue:** if every fiber tests fine but the link still won't come up, it's almost always a send/receive (polarity) mix-up — not a broken cable.

DO THIS — Don't panic about polarity

For now, just follow the work order. If it says ROLL, roll it. If the link won't come up and the fibers are clean and good, tell your lead 'I think it's a polarity issue' — they'll be impressed you spotted it.

STEP 5

Keep It Clean

This is the most important habit you will ever build as a fiber tech. Say it with us: **clean before you connect — every time.**

STOP — One speck of dust = a dead link

A fiber core is thinner than a hair. A tiny speck of dust sitting on it blocks the light — or gets ground in and **permanently damages** the end when you plug in. Dirt is the #1 cause of fiber problems, by far.

The three magic words: Inspect, Clean, Inspect

- 1 **Inspect** — look at the end with a fiber scope (a little microscope). Dirty?
- 2 **Clean** — use a one-click cleaner (dry). Still dirty? Use the wet-then-dry method.
- 3 **Inspect again** — only plug it in when it looks clean. If it won't clean up, it might be damaged — set it aside and tell your lead.

TIP — Clean both sides

Clean the **connector** AND the **port/bulkhead** you're plugging into. Dirt hides on both. And keep a dust cap on anything that isn't plugged in right now.

STEP 6

Read a Work Order

A work order (also called a build record) tells you exactly what to do. It has two ends — an **A-side** (start) and a **Z-side** (finish). Read it slowly, top to bottom.

The fields you'll actually use

What it says	What it means for you
Cable ID	The name of the cable. It should match the label on the cable.
A-side	Where you start: room, rack, device, and port.
Z-side	Where you finish: the other room, rack, device, and port.
Rack	Which cabinet to walk to.
Port / Adapter	The exact hole you plug into.
Strand	Which fiber to use. Wrong strand = no light.
Notes	Read this! It's where words like 'ROLL' or 'clean both ends' live.

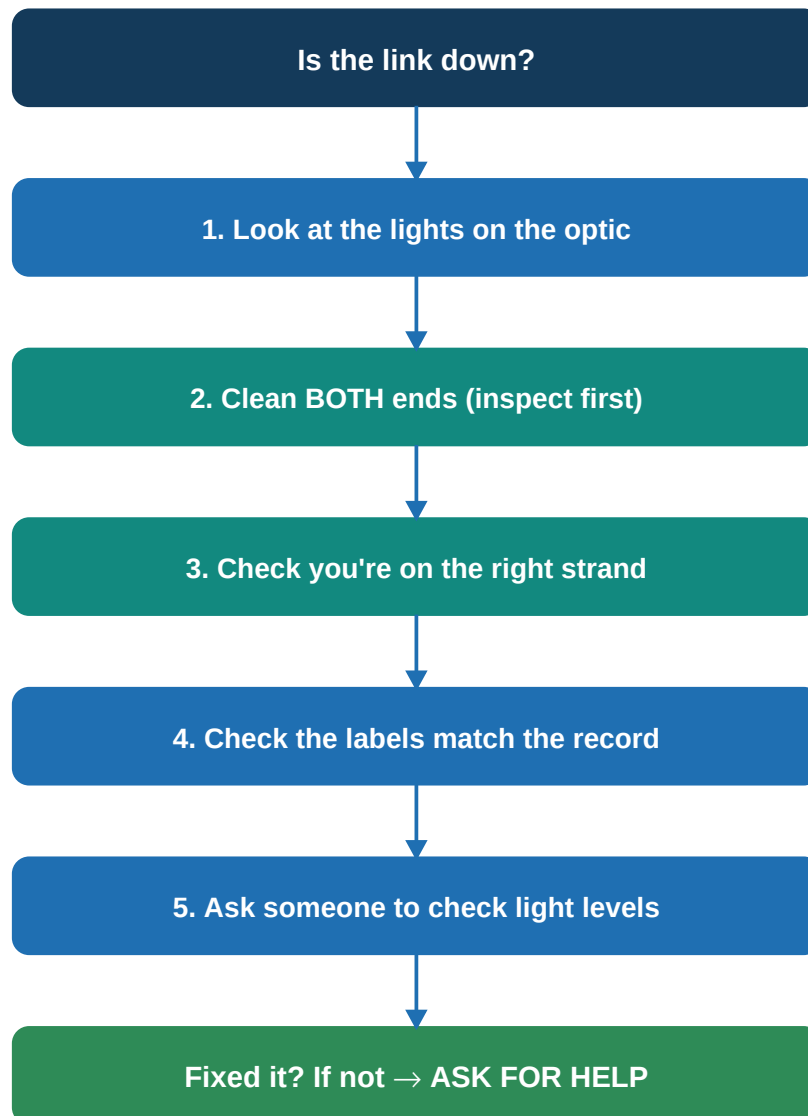
DO THIS — Simple habit: walk it, match it, then touch it

1) **Walk** to the rack. 2) **Match** the device, port, strand, and cable label to the record. 3) Only **then** touch anything. If something doesn't match — wrong label, no free port, colors don't match — take a photo and ask. Don't force it.

STEP 7

When a Link Is Down

A cable isn't working? Don't guess randomly. Work down this list in order — easy and cheap fixes first. Follow the arrows.

**TIP — Change ONE thing at a time**

Fix one thing, then check if it worked. If you swap the cable, the connector, and re-patch all at once, you'll never know what the problem was — and you might add a new one.

STEP 8

Labeling

A cable nobody can identify is a cable nobody can safely touch. Good labels keep the whole floor working. You'll often use a **Brady M511** label printer with a phone app.

The simple version

- 1 Turn on the printer, open the **Brady Express Labels** app, connect over Bluetooth.
- 2 Pick the **flag** label design.
- 3 Type the **Cable ID from the work order** (only add more if your site says to).
- 4 Check the preview, print a test — is it dark, centered, not cut off?
- 5 Wipe the cable clean, wrap the flag on (not over a connector or a bend), press it down.
- 6 Make sure it reads correctly. Label **both ends** of the cable.

DO THIS — Copy the site's naming exactly

Don't invent your own label format. Match how your site names things, letter for letter. That's what makes a label useful at 2 a.m. during an outage.

WORDS

Words You'll Hear

You don't need to memorize these — just know where to look. Here are the ones that come up most in your first weeks.

TX	Transmit — light going out (sending)	Patch / Jumper	The cable you plug in to make a connection
RX	Receive — light coming in	ODF	The frame/cabinet where fibers are patched
LC	The small, common connector (usually in pairs)	Rack / Cabinet	The metal enclosure that holds the gear
MPO / MTP	The wide connector with many fibers	U (rack unit)	A slot height in a rack, counted from the bottom
SMF	Single-mode fiber — goes far (yellow)	Optic / SFP	The little part that turns electricity into light
MMF	Multimode fiber — shorter runs (aqua/orange)	Dust cap	The cap that keeps a connector clean when unplugged
APC	Green connector polish — only mates with green	One-click cleaner	The pen tool you clean connectors with
UPC	Blue connector polish — only mates with blue	Scope	The little microscope for looking at fiber ends
Strand	One single fiber inside a cable	LOS	Loss of Signal — no light is getting through
ROLL	Swap the two fibers so send meets receive	Bend radius	How tight you're allowed to bend a cable (don't kink it)
Polarity	Making sure TX reaches RX (the crossover)	Cold aisle / Hot aisle	The cool front of racks / the warm back
A-side / Z-side	The start end / the finish end of a link	NOC	Network Operations Center — the folks who watch the network

The 8 Rules to Remember

1

Never look into a fiber or a port. Use a scope.

2

Clean before you connect. Inspect, clean, inspect — every time.

3

Push until it clicks. Loose looks connected but isn't.

4

Don't bend it tight or crush it. No loops smaller than a soda can.

5

Match the colors. Green to green, blue to blue — never mix.

6

The record is the boss. If it doesn't match, stop and ask.

7

Label both ends, exactly how your site names things.

8

Change one thing at a time when fixing a problem.

DO THIS — And the most important rule of all

When in doubt, stop and ask. Every good senior tech was new once. Asking a question is how you learn — and how you avoid taking a service down. You've got this.